

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: REFLECTION AND CONGRUENCE

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

You have now completed all eight lessons in the Reflection and Congruence unit. Your task is to write a Final Explanation that answers the driving question: "Why does a matatu's rear-view mirror show an image that is the same size and shape as the real object behind it — yet appears flipped — and how can we use this to prove that the original and its mirror image are always congruent?" Use your Summary Table, your Cartesian plane coordinate work, your flag-construction activity, your congruence-test investigations, and any notes from your Driving Question Board to support your answer. Your response must connect every section back to the matatu mirror phenomenon on the Nairobi–Thika highway. Include specific mathematical evidence: coordinates, distances, angles, and congruence test names (SSS, SAS, AAS, RHS). Use Kenyan contexts and examples where possible (e.g., the lorry's headlights, the matatu conductor's rectangular mirror, shapes found in Kenyan cultural objects). Your complete Final Explanation must be a minimum of 300 words across all sections. Write clearly and precisely — imagine you are explaining this mystery to a classmate who missed the entire unit.

SECTION 1: THE MATATU MIRROR PHENOMENON — WHAT DID WE OBSERVE?

Describe the anchoring phenomenon in your own words. What exactly did the matatu conductor's mirror do to the image of the lorry behind the vehicle? Identify at least THREE specific observations (size, distance, orientation) that made this situation puzzling or interesting. Explain why these observations raised a mathematical question worth investigating.

On the Nairobi–Thika highway, a matatu conductor held up a small rectangular mirror so the driver could see vehicles approaching from behind. When a lorry appeared in the mirror, three things were immediately noticeable. First, the image of the lorry looked exactly the same size as the real lorry — it had not been shrunk or enlarged. Second, the lorry's image appeared to sit as far behind the mirror surface as the real lorry was in front of it — the distances were equal. Third, and most puzzling, the lorry's right headlight appeared on the LEFT side in the mirror, and its left headlight appeared on the RIGHT — the image was laterally flipped, or reversed left-to-right.

This created a genuine mathematical mystery. If the image is flipped, you might expect it to look different — like a distorted or stretched copy. But it did not look different at all in terms of size or shape. Every measurable feature of the lorry — the width of the grille, the height of the cab, the angles of its windscreen — appeared identical in the mirror. So the question became: how can something be 'flipped' and yet remain a perfect copy? This is exactly the puzzle that the mathematics of reflection and congruence helps us solve.

SECTION 2: THE FOUR PROPERTIES OF REFLECTION AND THE MIRROR LINE

Explain the four key properties of reflection that you discovered during the unit. For each property, use precise mathematical language and connect it back to the matatu mirror. Then explain how you can find the equation of the mirror line when you are given an object point and its image point on the Cartesian plane. Include a specific coordinate example in your explanation.	Through our investigations on the Cartesian plane, we established four properties that every reflection obeys. First, the mirror line is the perpendicular bisector of the line segment joining any object point to its image point — meaning it cuts that segment at exactly 90° and at its exact midpoint. Second, every object point and its image point are equidistant from the mirror line — the lorry is, say, 30 metres in front of the mirror surface, and its image appears exactly 30 metres behind it. Third, points that lie ON the mirror line do not move — they map to themselves. Fourth, the mirror line preserves all lengths and all angles in the figure, which is why the image looks identical in size and shape to the original. To find the mirror line from coordinates, consider an object point $A(2, 6)$ and its image $A'(6, 2)$. The midpoint M of AA' is $((2+6)/2, (6+2)/2) = (4, 4)$. The gradient of AA' is $(2-6)/(6-2) = -1$, so the mirror line, being perpendicular to AA', has gradient $+1$. Using point-gradient form through $M(4, 4)$: $y - 4 = 1(x - 4)$, which simplifies to $y = x$. This tells us the mirror line is the line $y = x$ — which we can verify because reflecting any point over $y = x$ swaps its coordinates, exactly as the lorry's left and right sides are swapped in the matatu's mirror. The mirror line is the mathematical 'hinge' of the entire phenomenon.
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SECTION 3: WHY THE IMAGE IS FLIPPED — LATERAL REVERSAL EXPLAINED

Explain in mathematical terms why a reflected image appears 'flipped' or laterally reversed. Use the idea of orientation (the order in which vertices appear — clockwise vs. anti-clockwise) to distinguish between direct congruence and opposite (indirect) congruence. Connect this to the paper-template triangle activity: why could you sometimes flip one triangle perfectly onto the other, and why did this sometimes require lifting the triangle off the page?	When you reflect a shape, the image is a mirror copy — every length and every angle is preserved, but the ORDER in which the vertices appear around the shape is reversed. Mathematicians call this a change in orientation. If the vertices of the original lorry outline are labelled A, B, C going CLOCKWISE, then the vertices of the mirror image go ANTI-CLOCKWISE in the same order. This left-to-right flip is called lateral reversal, and it is the reason the lorry's right headlight appears on the left in the matatu's mirror. In our paper-template triangle activity, we cut two identical triangles from the same template — one representing the 'object' lorry and one representing the 'mirror image.' When we tried to place one on top of the other, we found that for a scalene triangle (one with no lines of symmetry), we could NOT slide or rotate the image triangle to fit the object triangle while keeping both flat on the page. We had to LIFT the image triangle, flip it over, and then it matched perfectly. This is called opposite congruence (or indirect congruence) — the two shapes are congruent, but you need a flip (reflection) to move from one to the other. By contrast, two shapes that can be made to coincide using only slides and rotations (no flip) are called directly congruent. The matatu mirror always produces opposite congruence — the image is a perfect copy that is also a mirror twin.
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SECTION 4: CONGRUENCE TESTS — PROVING THE MIRROR IMAGE IS ALWAYS AN EXACT COPY

Explain what congruence means and why it matters in the context of the matatu mirror. Describe ALL FOUR triangle congruence tests (SSS, SAS, AAS, RHS) using clear definitions. Then choose ONE test and use it to formally prove that a triangle in the original lorry outline and its reflected	Two shapes are congruent if they have exactly the same shape and exactly the same size — every corresponding side is equal in length and every corresponding angle is equal in measure. In the context of the matatu mirror, congruence answers the key question: even though the lorry's image is laterally flipped, is it truly an identical copy? The four congruence tests for triangles give us a way to answer YES with mathematical certainty, without needing to measure all six parts of both triangles. The four tests are: SSS (Side-Side-Side) — if all three sides of one triangle equal all three sides of another, the triangles are congruent; SAS (Side-Angle-Side) — if two sides and the included angle of one triangle equal the corresponding two sides and included angle of another, they are congruent; AAS (Angle-Angle-Side) — if two angles and a non-included side of one triangle equal the corresponding parts of another, they are congruent; and RHS (Right angle-Hypotenuse-Side) — if both triangles are right-angled and share the same hypotenuse and one other side, they are congruent.
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image are congruent. Use specific side and angle labels in your proof.	Using SSS to prove the mirror image is congruent: Consider a triangle PQR on the lorry's grille outline with P(1,1), Q(4,1), R(4,5). Reflecting over the y-axis gives image triangle P'Q'R' with P'(-1,1), Q'(-4,1), R'(-4,5). By the distance formula: PQ = 3 units = P'Q'; QR = 4 units = Q'R'; PR = 5 units = P'R'. Since all three pairs of corresponding sides are equal, triangle PQR $\cong$ triangle P'Q'R' by SSS. The matatu mirror always produces a congruent image because reflection preserves ALL distances — this is guaranteed by the four properties of reflection we established in Section 2.
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SECTION 5: ANSWERING THE DRIVING QUESTION — THE FULL MATATU MIRROR EXPLANATION

Now bring everything together to fully answer the driving question. Explain in a unified argument: (a) why the mirror image is the same size and shape as the real object, (b) why it appears flipped, and (c) how the congruence tests prove the original and its mirror image are always congruent. Reflect on what was most surprising to you in this unit and describe one place — other than a matatu mirror — where you see reflection and congruence working together in everyday Kenyan life.	The matatu mirror mystery is now fully solved. The image of the lorry in the conductor's mirror is the same size and shape as the real lorry because reflection is an isometry — a transformation that preserves all distances and all angles. Every point on the lorry maps to an image point that is the same distance from the mirror line, on the opposite side, along a perpendicular. Because no stretching or squashing occurs, every length and every angle in the image equals the corresponding length or angle in the original. This is why the lorry's cab looks just as wide and just as tall in the mirror as it does in real life. The image appears flipped because reflection reverses orientation. The vertices of the lorry's outline, which go clockwise in reality, go anti-clockwise in the mirror. This is lateral reversal — it affects LEFT and RIGHT but not UP and DOWN in a standard rear-view mirror. It is the mathematical reason why, if the lorry had a sign reading 'THIKA,' the mirror would show it as reversed. Despite the flip, we can PROVE the original and image are always congruent using the SSS, SAS, AAS, or RHS tests. Since reflection preserves all side lengths, SSS alone guarantees congruence for any triangle we choose to isolate from the lorry's outline. The shapes are congruent — they are just oppositely oriented, making this a case of indirect (opposite) congruence. The most surprising insight from this unit was that 'flipped' does not mean 'different.' A shape and its mirror image are as identical as two mandazi cut from the same dough cutter — they match in every measurement, even if one is face-up and the other face-down. Beyond the matatu, you can see reflection and congruence in the symmetrical patterns on a Kikoi fabric sold at Mombasa market, in the two identical wings of a butterfly photographed at Lake Nakuru, or in the congruent triangular roof trusses of houses being built in Kisumu — every pair of trusses must be congruent to ensure the roof sits level and strong.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explanation of Reflection Properties	Excellent (4): Accurately states and explains all four properties of reflection (perpendicular bisector, equal distances, invariant points, preservation of lengths and angles) using precise mathematical language. Connects every property explicitly to the matatu mirror phenomenon with correct coordinate or geometric evidence.	Proficient (3): Accurately states and explains at least three of the four properties of reflection. Uses mostly correct mathematical language and makes a clear connection to the matatu mirror phenomenon, with some coordinate or geometric evidence provided.	Developing (1–2): States one or two properties of reflection, often imprecisely or incompletely. Mathematical language is vague or contains errors. The connection to the matatu mirror phenomenon is absent or superficial, and little or no coordinate evidence is included.

Understanding of Lateral Reversal and Orientation	Excellent (4): Clearly and accurately explains lateral reversal using the concept of orientation (clockwise vs. anti-clockwise vertex order). Correctly distinguishes between direct congruence and opposite (indirect) congruence. Uses the paper-template triangle activity or a specific shape example to illustrate why a flip is needed to superimpose a reflected image.	Proficient (3): Explains lateral reversal and correctly identifies that the image is flipped. References orientation or vertex order at least once. Mentions the distinction between direct and opposite congruence, though the explanation may lack full precision or a concrete example.	Developing (1–2): Describes the image as 'flipped' or 'reversed' but cannot explain this mathematically in terms of orientation. Does not distinguish between direct and opposite congruence, or confuses the two. Explanation is largely descriptive with no mathematical justification.
Application of Triangle Congruence Tests	Excellent (4): Correctly defines all four congruence tests (SSS, SAS, AAS, RHS) with accurate descriptions. Selects an appropriate test and constructs a complete, logically structured formal proof using specific labelled coordinates or measurements that demonstrates the original triangle and its reflected image are congruent. Conclusion is explicitly linked back to the driving question.	Proficient (3): Correctly defines at least three of the four congruence tests. Attempts a formal proof using one test with specific labels or measurements; the proof is mostly correct but may have one minor error or omission. Makes a reasonable link between the proof and the matatu mirror context.	Developing (1–2): Names one or two congruence tests but definitions are incomplete or contain errors. Does not attempt a formal proof, or the attempted proof is missing labels, measurements, or a logical structure. The connection between congruence tests and the mirror phenomenon is unclear or absent.
Finding the Mirror Line (Equation)	Excellent (4): Accurately explains the method for finding the mirror line — using the midpoint of the object-image segment and the perpendicular gradient — and demonstrates the method with a fully worked coordinate example leading to a correct equation. Clearly explains what the equation tells us about the geometry of the situation.	Proficient (3): Explains the method for finding the mirror line with mostly correct steps and includes a coordinate example. The final equation may contain a minor arithmetic error, but the method and reasoning are sound. Some explanation of what the equation represents is provided.	Developing (1–2): Attempts to find the mirror line but the method is incomplete or incorrect (e.g., finds midpoint but does not use perpendicular gradient, or uses an incorrect formula). The coordinate example is missing or contains significant errors. Little or no explanation of the meaning of the equation is given.
Coherence, Kenyan Context, and Driving Question Response	Excellent (4): The Final Explanation is well-organised, clearly written, and meets the 300-word minimum. Every section connects back to the matatu mirror phenomenon in a meaningful way. At least two specific Kenyan contexts (places, objects, foods, or cultural items) are used as examples. The driving question is answered completely and explicitly in Section 5, integrating evidence from all four content areas.	Proficient (3): The Final Explanation is mostly organised and mostly clear, meeting or approaching the 300-word minimum. Most sections connect to the matatu mirror phenomenon. At least one specific Kenyan context is used. The driving question is addressed in Section 5, though one content area may be insufficiently integrated.	Developing (1–2): The Final Explanation is below 300 words, poorly organised, or difficult to follow. Connections to the matatu mirror phenomenon are rare or generic. Kenyan contexts are absent or incorrectly applied. The driving question is only partially addressed or not addressed at all in Section 5.